Individual Case Safety Report (ICSR)

A complete pharmacovigilance assessment of one case, from receipt to reporting, in the role of a Drug Safety Associate

PHARMACOVIGILANCE INTERNSHIP | TASK 5

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Goal of this task: process one detailed case the way a Drug Safety Associate (DSA) would: identify the ADR, check the report is valid, enter and code it, assess causality, seriousness, severity and expectedness, decide the reporting pathway, and write the narrative. The case below is fictional and brings together everything from Tasks 1 to 4.

1. The ICSR and the Drug Safety Associate Role

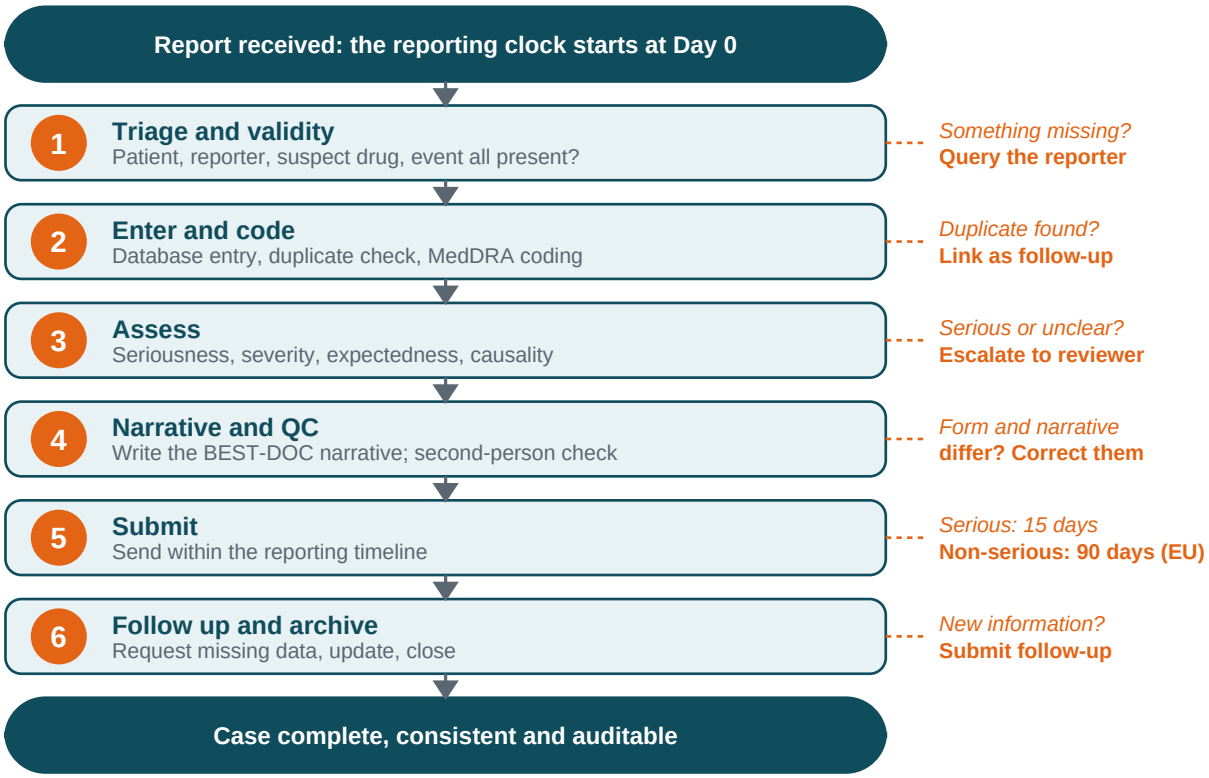
An **Individual Case Safety Report (ICSR)** is the structured record of one patient's suspected ADR. It is the unit of data that safety databases hold and regulators receive. The DSA receives raw reports, turns them into complete, coded, consistent ICSRs, and prepares them for medical review and submission.

ICSR block	What it holds	ICH E2B(R3) section
Case identification and source	Unique case ID, report type, dates of receipt, reporter and sender details	C
Patient	Initials or ID, age, sex, weight, medical history	D
Reaction / event	Coded terms, onset, outcome, seriousness criteria	E
Tests and procedures	Laboratory results with dates and reference ranges	F
Drug(s)	Role (suspect, interacting, concomitant), dose, dates, action taken, dechallenge	G
Narrative and assessment	Case summary, causality, sender's comments	H

Section letters follow E2B(R3); the older E2B(R2) format used sections A and B. Field names and screens vary between safety database systems.

2. DSA Workflow

Every case follows the same path. The numbers match the sections of this report.



3. The Case as Received

Initial report (received 20-Sep-2026, by email)

> *Subject: Possible drug reaction, statin and antibiotic.*
> Dear Drug Safety team, I am writing about my patient, a 67-year-old man (82 kg), who was admitted on 7 September with severe muscle pain, weakness and dark urine. He has taken atorvastatin 40 mg at night for about 14 months, plus amlodipine, metformin and aspirin. His GP started clarithromycin 500 mg twice a day for a chest infection on 1 September. The pain began on 5 September. CK was 18,400 and creatinine 2.1 (it was 1.0 in August). We stopped both drugs and gave IV fluids. He went home on the 15th with CK 480 and creatinine 1.1. I think the antibiotic and the statin interacted.
> *Dr. R. Mehta (example), Consultant Physician, City General Hospital (example)*

Follow-up 1 (received 05-Oct-2026): concomitant doses (amlodipine 5 mg daily, metformin 500 mg twice daily, aspirin 75 mg daily); metformin held in hospital and restarted 20-Sep-2026; no alcohol, grapefruit juice, trauma or strenuous exercise; TSH normal; on 29-Sep-2026 CK 160 U/L and creatinine 1.0 mg/dL; brand and batch numbers unknown.

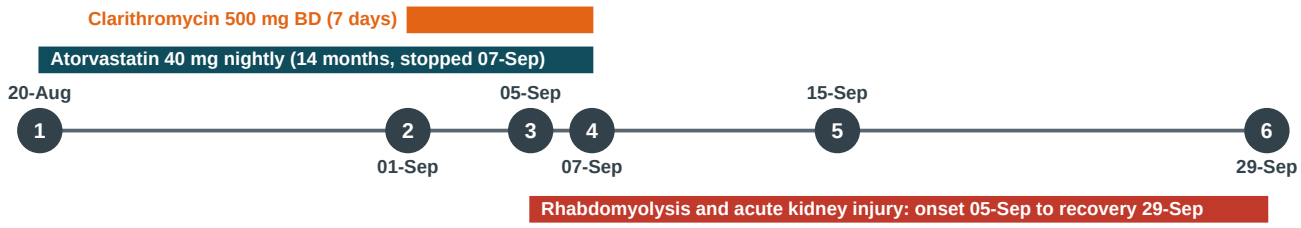
4. Step 1: ADR Identification and Validity

Minimum criterion	Present?	Evidence
Identifiable patient	**Yes**	67-year-old male, 82 kg.
Identifiable reporter	**Yes**	Dr. R. Mehta, Consultant Physician, City General Hospital.
Suspect drug	**Yes**	Atorvastatin; clarithromycin.
Adverse event	**Yes**	Rhabdomyolysis with acute kidney injury.

Valid ICSR. Day 0 = 20-Sep-2026 (the day the minimum information was first received).

Is it an ADR? Yes, a suspected one. The event followed the start of clarithromycin in a patient on long-term atorvastatin, rhabdomyolysis is a known statin effect, and the two drugs share a recognised interaction (clarithromycin inhibits CYP3A4, raising statin exposure). It is a **Type A** (augmented) reaction: predictable from the pharmacology. Special situations to flag are a **drug interaction** and a **possible medication error**, because the product information advises caution or a dose limit for this combination. Both go to the medical reviewer.

5. Step 2: Build the Timeline



#	Date	Day*	Event
1	20-Aug-2026	-	Baseline creatinine 1.0 mg/dL.
2	01-Sep-2026	1	Clarithromycin 500 mg twice daily started; atorvastatin continued.
3	05-Sep-2026	5	Thigh muscle pain, weakness and dark urine begin (4 days after clarithromycin started).
4	07-Sep-2026	7	Admitted: CK 18,400 U/L, creatinine 2.1 mg/dL. Atorvastatin and clarithromycin stopped.
5	15-Sep-2026	15	Discharged: CK 480 U/L, creatinine 1.1 mg/dL.
6	29-Sep-2026	29	CK 160 U/L, creatinine 1.0 mg/dL: recovered.

**Day 1 = first clarithromycin dose.*

6. Step 3: The Completed ICSR

Current version: **Follow-up 1**. Terms shown are as entered by the DSA; MedDRA coding is suggested and subject to review.

CASE IDENTIFICATION AND SOURCE	
Case ID (example)	XX-EXAMPLE-2026-000123 (version: Follow-up 1)
Report type / source	Spontaneous report from a healthcare professional (physician)
Country of occurrence	India (example)
Date of receipt	Initial: 20-Sep-2026 (Day 0). Follow-up 1: 05-Oct-2026.
Reporter	Dr. R. Mehta (example), Consultant Physician, City General Hospital (example)

PATIENT	
Initials / ID	R.K. / PT-5001
Age / Sex / Weight / Height	67 years / Male / 82 kg / UNK
Medical history	Hypertension; type 2 diabetes; hyperlipidaemia. No previous muscle problems on atorvastatin.
Habits and exposures	No alcohol; no grapefruit juice; no recent trauma or strenuous exercise.

Reaction / event

Reported term	MedDRA PT (suggested)	Onset	Outcome	Serious?
Rhabdomyolysis	Rhabdomyolysis	05-Sep-2026	Recovered 29-Sep-2026	**Yes**: hospitalisation
Kidney injury (creatinine 2.1 mg/dL)	Acute kidney injury	07-Sep-2026 (detected)	Recovered 29-Sep-2026	**Yes**: hospitalisation
Muscle pain and weakness	Myalgia; Muscular weakness	05-Sep-2026	Recovered	Part of the above
Dark urine	Chromaturia	05-Sep-2026	Recovered	Part of the above

Tests and procedures

Date	Test	Result	Reference
20-Aug-2026	Creatinine	1.0 mg/dL	Baseline
07-Sep-2026	Creatine kinase (CK)	**18,400 U/L**	Normal up to about 200
07-Sep-2026	Creatinine	**2.1 mg/dL**	Baseline 1.0
07-Sep-2026	Potassium	5.6 mmol/L	3.5 to 5.0
07-Sep-2026	ALT	112 U/L	Normal up to about 40
07-Sep-2026	Urine myoglobin	Positive	Negative
07-Sep-2026	TSH	Normal	Normal range
12-Sep-2026	CK / creatinine	2,100 U/L / 1.3 mg/dL	Falling
15-Sep-2026	CK / creatinine	480 U/L / 1.1 mg/dL	Discharge
29-Sep-2026	CK / creatinine	160 U/L / 1.0 mg/dL	Normal

Drug information

Role	Drug and dose	Indication	Dates	Action and dechallenge
Suspect	Atorvastatin 40 mg once nightly, oral	Hyperlipidaemia	Start about 14 months earlier (date UNK). Stop 07-Sep-2026.	Withdrawn. Dechallenge positive*. Rechallenge not done.
Interacting	Clarithromycin 500 mg twice daily, oral	Lower respiratory tract infection	01-Sep-2026 to 07-Sep-2026 (7 days).	Stopped (course complete). Dechallenge positive*.
Concomitant	Amlodipine 5 mg once daily, oral	Hypertension	Ongoing; start UNK.	Unchanged.
Concomitant	Metformin 500 mg twice daily, oral	Type 2 diabetes	Ongoing; held 07-Sep to 20-Sep-2026.	Held for kidney injury; restarted 20-Sep-2026.
Concomitant	Aspirin 75 mg once daily, oral	UNK	Ongoing; start UNK.	Unchanged.

**Both drugs were stopped together and IV fluids were given, so the dechallenge cannot separate the two drugs. Brand, batch and manufacturer: UNK.*

TREATMENT OF THE EVENT	
Treatment given	Intravenous saline hydration; potassium monitoring; metformin held. No dialysis. Hospital stay 07-Sep-2026 to 15-Sep-2026 (8 days).

7. Step 4: Causality Assessment

Primary suspect: **atorvastatin**. Clarithromycin is recorded as the **interacting drug** that raised atorvastatin exposure, so it is treated as part of the mechanism and not as an independent alternative cause.

#	Naranjo question	Answer	Score	Evidence
1	Previous conclusive reports?	Yes	+1	Statin rhabdomyolysis and this interaction are well documented.
2	Event after the drug was given?	Yes	+2	Onset Day 5 of clarithromycin, on long-term atorvastatin.
3	Improved on withdrawal?	Yes	+1	CK and creatinine fell after both drugs stopped (fluids too).
4	Recurred on re-exposure?	Not done	0	No rechallenge.
5	Alternative causes?	No	+2	No trauma, exercise or alcohol; TSH normal; interaction is the mechanism.
6	Reappeared with placebo?	Not done	0	-
7	Toxic drug level?	Unknown	0	Drug levels not measured.
8	Dose-response?	Unknown	0	No dose change data.
9	Similar reaction before?	No	0	Tolerated atorvastatin 40 mg for 14 months.
10	Objective evidence?	Yes	+1	CK 18,400 U/L, myoglobinuria, creatinine rise.

Drug	WHO-UMC	Naranjo	Reason
Atorvastatin (suspect)	**Probable / Likely**	**7 (Probable)**	Plausible timing, no other cause, positive dechallenge, labelled reaction. Not Certain: no rechallenge.
Clarithromycin (interacting)	**Probable / Likely**	-	Known CYP3A4 interaction; the event began 4 days after it was added to stable statin therapy.

For interactions the Drug Interaction Probability Scale (DIPS) is a further option; here the scales are applied to the primary suspect.

8. Step 5: Seriousness, Severity and Expectedness

Criterion (D-L-H-D-C-M)	Met?	Basis
Death, disability, congenital anomaly	No	Full recovery; no birth defect involved.
Life-threatening	No	Not judged to be at immediate risk of death (no intensive care, no dialysis).
Hospitalisation	**Yes**	Admitted 07-Sep-2026, discharged 15-Sep-2026 (8 days).
Medically important	Flag	Kidney injury with markedly raised CK: DSA flags it; the medical reviewer confirms.

Dimension	Result and basis
Seriousness	**Serious**: hospitalisation.
Severity	**Moderate**: Hartwig Level 4(b), the ADR was the reason for admission. No intensive care or permanent harm.
Expectedness	**Expected**: product information describes rhabdomyolysis with renal failure and the clarithromycin interaction. The DSA verifies this against the reference safety information version in force.
ADR type	Type A (augmented): predictable and exposure-related.
Preventability	Likely preventable: the interaction is labelled, so a non-interacting antibiotic or a statin pause could have been chosen.
Reporting category	**Serious + expected + related**: expedited ICSR within 15 calendar days of Day 0.
Clinical impact	**Medium to high**: full recovery, but kidney injury and an 8-day admission from a known, avoidable interaction. Worth tracking in aggregate statin-macrolide data.

9. Step 6: Reporting Decision and Case Clock

Item	Date	Note
Valid initial report received (Day 0)	20-Sep-2026	Clock starts.
Initial ICSR due (Day 0 + 15 days)	**05-Oct-2026**	Serious case; expedited.
Follow-up 1 received	05-Oct-2026	Adds outcome, doses and exposure details.
Follow-up ICSR due (receipt + 15 days)	**20-Oct-2026**	Significant follow-up restarts the clock.

EU-style 15-day rule shown. In the US a serious, expected case usually goes in a periodic report. Follow your own SOP.

Flags for the medical reviewer

(1) Drug interaction with a labelled warning. (2) Possible medication error: atorvastatin 40 mg was continued with clarithromycin. (3) Confirm whether the kidney injury meets the medically important criterion. (4) Confirm expectedness against the current reference safety information.

10. Step 7: Case Narrative (BEST-DOC)

This spontaneous report from a physician concerns a 67-year-old male patient (82 kg) with hypertension, type 2 diabetes and hyperlipidaemia who developed rhabdomyolysis with acute kidney injury after clarithromycin was added to long-term atorvastatin. Concomitant medicines were amlodipine 5 mg daily, metformin 500 mg twice daily and aspirin 75 mg daily. He had taken atorvastatin 40 mg nightly for about 14 months without problems. On 01-Sep-2026 clarithromycin 500 mg twice daily was started for a lower respiratory tract infection (7-day course) and atorvastatin was continued. On 05-Sep-2026 (Day 5) he developed severe thigh muscle pain, weakness and dark urine. He had no trauma, strenuous exercise, alcohol or grapefruit juice intake. He was admitted on 07-Sep-2026. CK was 18,400 U/L (normal up to about 200), creatinine 2.1 mg/dL (1.0 mg/dL on 20-Aug-2026), potassium 5.6 mmol/L, ALT 112 U/L and urine myoglobin positive; TSH was normal. Rhabdomyolysis with acute kidney injury was diagnosed. Atorvastatin and clarithromycin were stopped on 07-Sep-2026, metformin was held, and he received intravenous saline hydration with potassium monitoring. Dialysis was not needed. By 12-Sep-2026 CK was 2,100 U/L and creatinine 1.3 mg/dL. He was discharged on 15-Sep-2026 (CK 480 U/L, creatinine 1.1 mg/dL). On 29-Sep-2026 CK was 160 U/L and creatinine 1.0 mg/dL, and the events were considered recovered. No rechallenge was done. The reporter suspects an interaction between atorvastatin and clarithromycin. **Company assessment:** serious (hospitalisation), moderate (Hartwig Level 4), expected; causality for atorvastatin is Probable (WHO-UMC; Naranjo 7), with clarithromycin as an interacting drug.

11. Step 8: Quality Check and Follow-up

Quality check	Done
Valid report; Day 0 and due dates recorded; no duplicate, follow-up linked to the same case	[ ]
Dates in one format; onset is after the suspect drug start; timeline matches the narrative	[ ]
Suspect, interacting and concomitant drugs separated; dose, route, dates and action recorded	[ ]
Events and drugs coded; diagnosis coded with supporting symptoms	[ ]
Seriousness criteria, severity scale, expectedness basis and causality method all stated	[ ]
Narrative matches every field on the form; missing data written as UNK, never guessed	[ ]
Flags raised to the medical reviewer	[ ]

Open follow-up items

- Brand, batch and manufacturer of both drugs (currently UNK); exact atorvastatin start date.
- Plan for statin therapy after recovery, and why the combination was prescribed (supports the medication-error review).

This case is fictional and for training only. For real cases follow your organisation's SOPs and current guidance (for example ICH E2A, E2B(R3), E2D, CIOMS and local regulations).